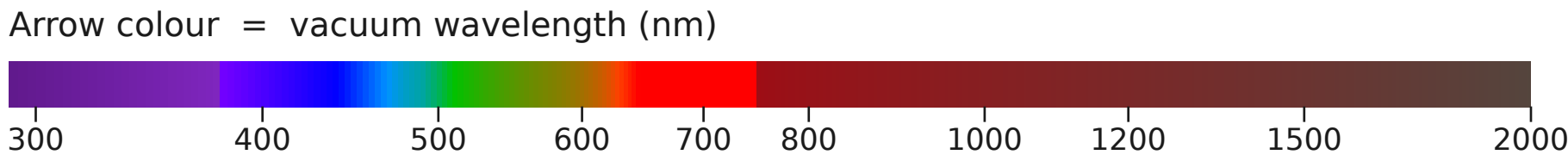


Re (Re I). Strongest classified transitions below 2 μm (NIST): 67 levels, 90 lines.

Re⁺ ionisation limit 63181.60 cm⁻¹ = 7.83352 eV (158.274 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U | \text{er} | U \rangle$ in ϵ_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



odd

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \mathbf{e} J\rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$a^5S_{1/2} - 498\frac{1}{2}$	200.4186	200.3538	1495.8316	1.070	7.210e+07	-	-	-
$a^5S_{1/2} - 4954\frac{1}{2}$	201.8532	201.7881	1485.2006	0.925	3.510e+07	-	-	-
$a^5S_{1/2} - 487\frac{1}{2}$	204.9754	204.9097	1462.5780	1.364	5.470e+07	-	-	-
$a^5S_{1/2} - 467\frac{3}{2}$	213.9798	213.9124	1401.0315	0.811*	3.400e+07	8.842	61.1	-
$a^5S_{1/2} - 461\frac{1}{2}$	216.8622	216.7941	1382.4102	1.237*	5.070e+07	15.3	52.6	-
$a^5S_{1/2} - 459\frac{1}{2}$	217.6886	217.6204	1377.1620	1.530*	5.750e+07	12.63	72.4	-
$a^5S_{1/2} - 438\frac{3}{2}$	228.2323	228.1619	1313.5410	1.267*	3.420e+07	8.421	64.8	-
$a^5S_{1/2} - x^6P^o_{8/2}$	228.8214	228.7509	1310.1590	1.119	5.290e+07	-	-	-
$a^5S_{1/2} - x^6P^o_{6/2}$	229.5191	229.4484	1306.1765	1.107	3.420e+07	-	-	-
$a^5S_{1/2} - y^6D^o_{5/2}$	242.0544	241.9809	1238.5332	1.190	3.370e+07	-	-	-
$a^5S_{1/2} - 411\frac{1}{2}$	242.9312	242.8575	1223.0630	1.764	5.500e+07	-	-	-
$a^5S_{1/2} - 408\frac{3}{2}$	245.0449	244.9707	1223.4185	0.963	3.190e+07	-	-	-
$a^5S_{1/2} - x^6P^o_{7/2}$	250.9741	250.8985	1194.5156	1.526	3.730e+07	-	-	-
$a^5D_{1/2} - 466\frac{1}{2}$	308.3321	308.2426	972.3037	1.622*	3.030e+07	10.14	47.6	-
$a^5D_{1/2} - 439\frac{1}{2}$	336.3253	336.2287	891.3765	2.061*	3.770e+07	16.08	37.3	-
$a^5S_{1/2} - z^6P^o_{7/2}$	346.1439	346.0484	866.0920	2.282	3.180e+07	-	-	-

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$